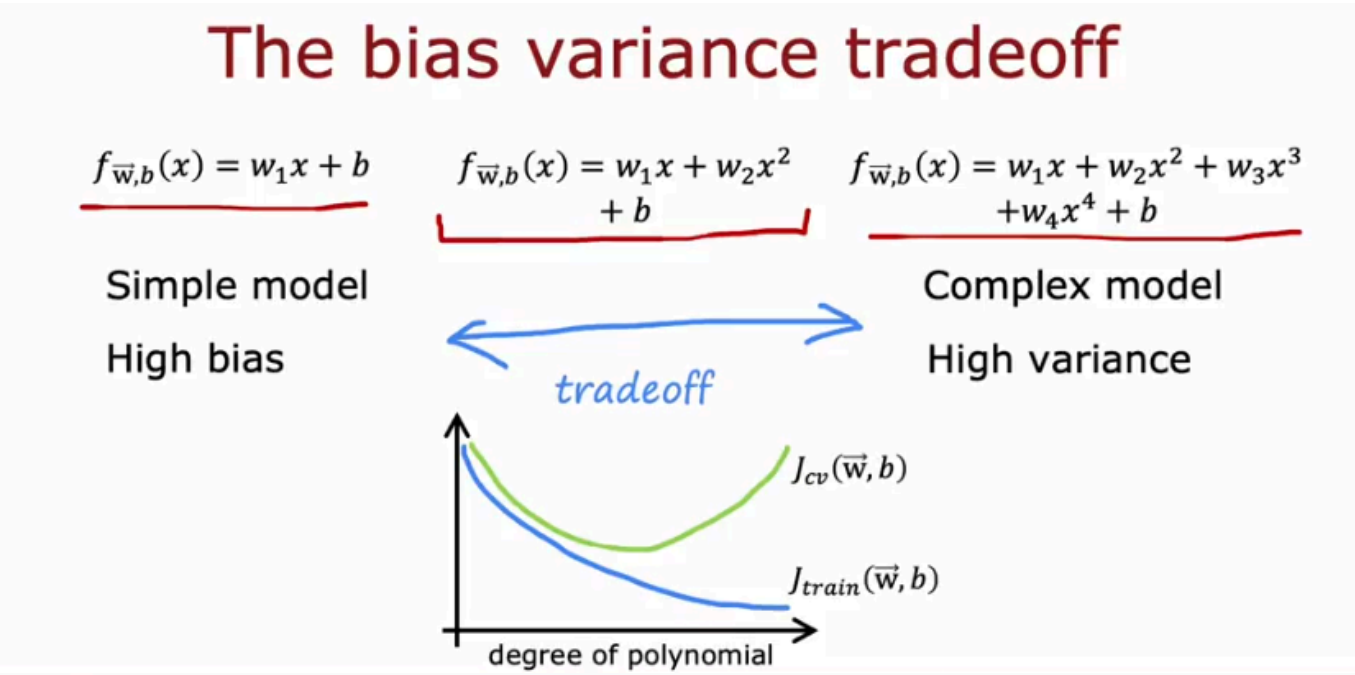
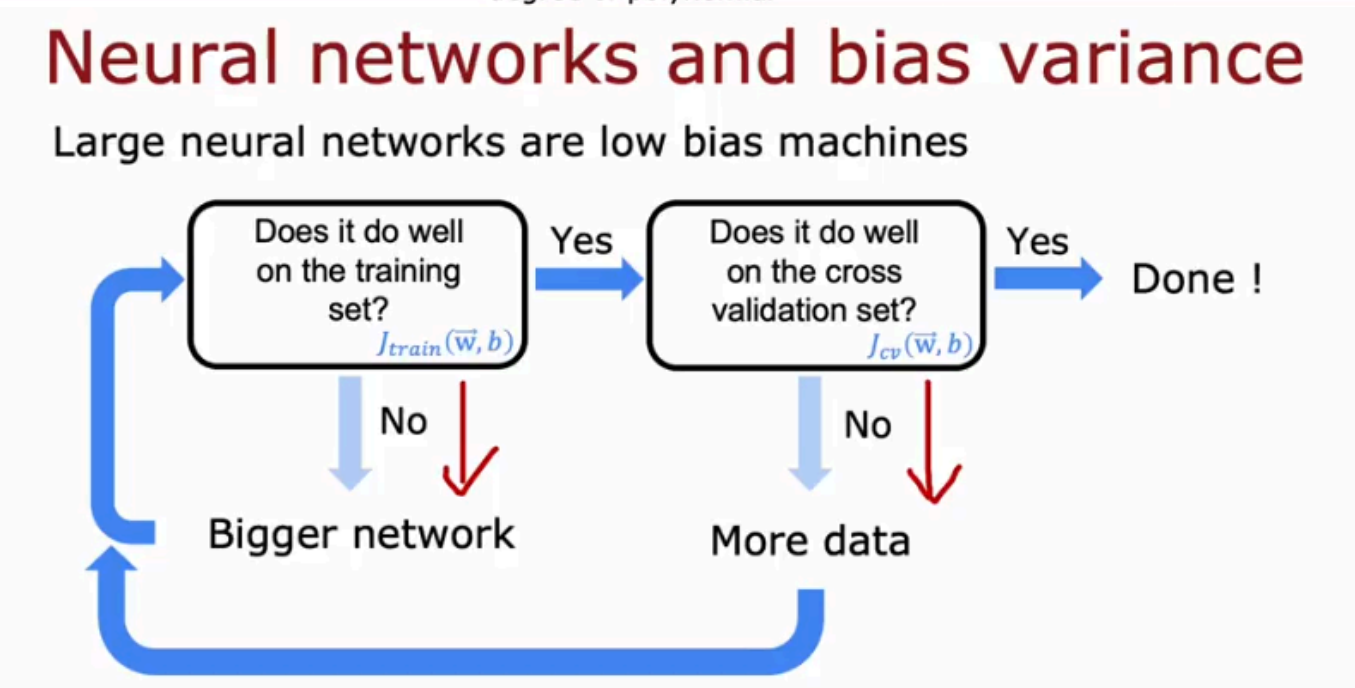
Day 22. Machine learning develptment process

Continue on bias/variance and neural networks

> One of the reasons that neural networks have been so successful is because your networks, together with the idea of big data or hopefully having large data sets.





Neural network regularization

$$J(\mathbf{W},\mathbf{B}) = \frac{1}{m}\sum_{i=1}^{m} L\left(f\left(\vec{x}^{(i)}\right), y^{(i)}\right) + \frac{\lambda}{2m}\sum_{\text{all weights } \mathbf{W}} \left(w^2\right)$$

b

Unregularized MNIST model

```
layer_1 = Dense(units=25, activation="relu")
layer_2 = Dense(units=15, activation="relu")
layer_3 = Dense(units=1, activation="sigmoid")
model = Sequential([layer_1, layer_2, layer_3])
```

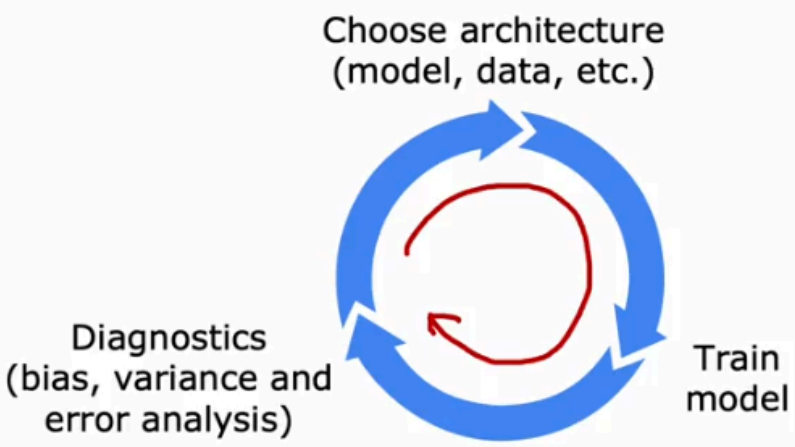
Regularized MNIST model

```
layer_1 = Dense(units=25, activation="relu", kernel_regularizer=L2(0.01))
layer_2 = Dense(units=15, activation="relu", kernel_regularizer=L2(0.01))
layer_3 = Dense(units=1, activation="sigmoid", kernel_regularizer=L2(0.01))
model = Sequential([layer_1, layer_2, layer_3])
```

Machine learning development process

Iterative loop of ML development

Iterative loop of ML development



Spam classification example

```
From: cheapsales@buystufffromme.com
To: Andrew Ng
Subject: Buy now!

Deal of the week! Buy now!
Rolex w4tchs - $100
Med1cine (any kind) - £50
Also low cost M0rgages
available.
```

```
From: Alfred Ng
To: Andrew Ng
Subject: Christmas dates?

Hey Andrew,
Was talking to Mom about plans
for Xmas. When do you get off
work. Meet Dec 22?
Alf
```

Building a spam classifier

Supervised learning: $\vec{x}$ = features of email
y = spam (1) or not spam (0)

Features: list the top 10,000 words to compute $x_1, x_2, \cdots, x_{10,000}$

$$\vec{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 0 \\ \vdots \end{bmatrix} \begin{matrix} a \\ andrew \\ buy \\ deal \\ discount \\ \vdots \end{matrix}$$

```
From: cheapsales@buystufffromme.com
To: Andrew Ng
Subject: Buy now!

Deal of the week! Buy now!
Rolex w4tchs - $100
Med1cine (any kind) - £50
Also low cost M0rgages
available.
```

Building a spam classifier

How to try to reduce your spam classifier's error?

- Collect more data. E.g., "Honeypot" project.
- Develop sophisticated features based on email routing (from email header).
- Define sophisticated features from email body. E.g., should "discounting" and "discount" be treated as the same word.
- Design algorithms to detect misspellings. E.g., w4tches, med1cine, m0rtgage.

Error analysis

Error analysis

m_{cv}= 500 examples in cross validation set.

Algorithm misclassifies 100 of them.

Manually examine 100 examples and categorize them based on common traits.

Pharma:
Deliberate misspellings (w4tches, med1cine):
Unusual email routing:
Steal passwords (phishing):
Spam message in embedded image:

Error analysis

m_{cv}= ~~500~~ 5000 examples in cross validation set.

Algorithm misclassifies ~~100~~ 1000 of them.

Manually examine 100 examples and categorize them based on common traits.

more data
features

Pharma: 21
Deliberate misspellings (w4tches, med1cine): 3
Unusual email routing: 7
Steal passwords (phishing): 18
Spam message in embedded image: 5

Adding data

Adding data

Add more data of everything. E.g., "Honeypot" project.

Add more data of the types where error analysis has indicated it might help.

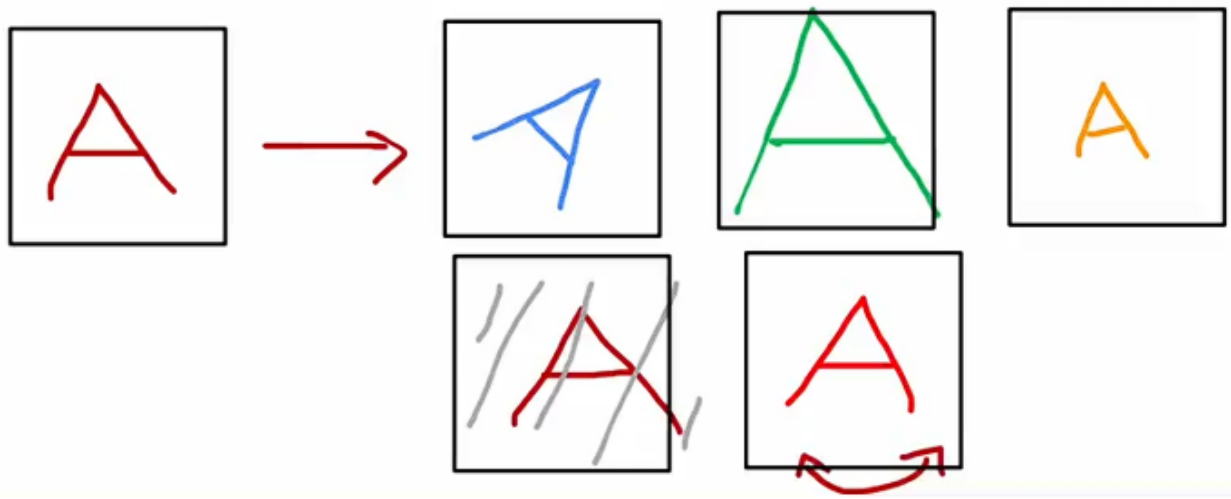
Pharma spam

E.g., Go to unlabeled data and find more examples of Pharma related spam.

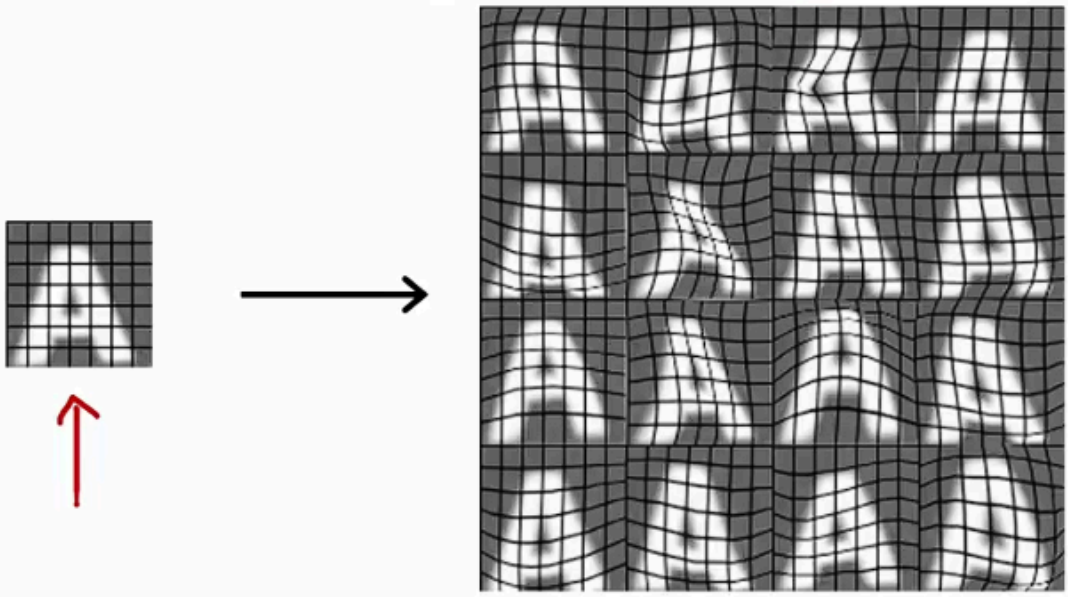
Beyond getting brand new training examples (x,y), another technique: Data augmentation

Data augmentation

augmentation means modifying and existing training example to create a new training example



Data augmentation by introducing distortions



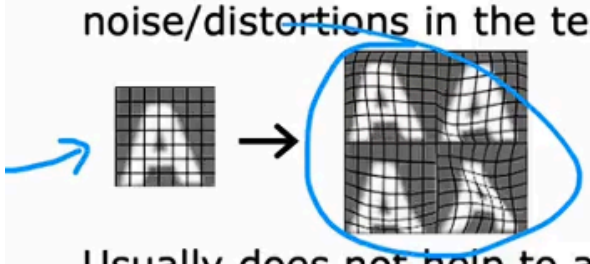
Data augmentation for speech

Speech recognition example

- Original audio (voice search: "What is today's weather?")
- + Noisy background: Crowd
- + Noisy background: Car
- + Audio on bad cellphone connection

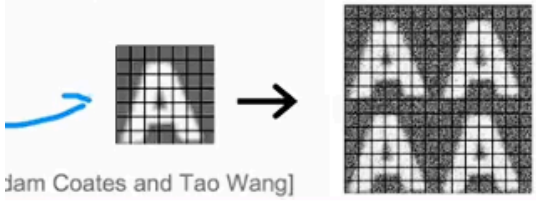
Data augmentation by introducing distortions

Distortion introduced should be representation of the type of noise/distortions in the test set.



Audio:
Background noise,
bad cellphone connection

Usually does not help to add purely random/meaningless noise to your data.



dam Coates and Tao Wang]

x_i=intensity (brightness) of pixel i

$x_i \leftarrow x_i +$ random noise

Data synthesis

Synthesis: using artificial data inputs to create a new training example.

Artificial data synthesis for photo OCR

Real data

Synthetic data

Engineering the data used by your system

Conventional model-centric approach:

AI = Code + Data
(algorithm/model)

work on this

Data-centric approach:

AI = Code + Data
(algorithm/model)

work on this